

CSE 331/EEE453: Microprocessor Interfacing and Embedded Systems

Assignment#2

Summer 2025

Name: _____

Student ID: _____

Direction: You can consult any resources such as books, online references, and videos for this assignment, however, you have to properly cite and paraphrase your answers when it is necessary. There will be points for partial attempts. There will be three days of grace period for late submission, however, 20% of points will be deducted.

Submission guidelines: All our assignments will be based on 100 points so that we can assign weights to get your points for final grade. Whoever fails to submit within the time period assigned but submit the assignment within the next three days will be penalized by 20% deduction of points (e.g. start with -20 points within 100 points). The late submission will be open until semester to TA. After the first 1-3 days of grace period, the deduction will be 30%.

Note: You are allowed to use only instructions implemented by the actual ARM hardware. Note, base ten numbers are listed as normal (e.g. 23), binary numbers are prefixed with 0b or format such as XX_2 and hexadecimal numbers are prefixed with 0x or format such as XX_{16}/XX_{HEX}

Problem (10x10=100 points): Answer the following 10 questions:

- (10 points)** Write an assembly code for the following C code

```
int main (void)
{ int x=3; int y=5; int z=1;
  int total=0;
  total=x+y*z;
}
```
- (10 points)** What is pipelining? What do you think are the advantages of pipelining? How many pipeline stages does ARM Cortex-M4 have? Why does it have XX stages of pipeline?
- (10 points)** What is the purpose of a stack pointer (SP) and heap? Where is the address of the stack pointer stored? When to use stack and heap, and why?

4. **(10 points)** If you are given a 128 bit microprocessor? How many memory locations can it address? Why do most microcontrollers have 32 bit architecture rather than 128 bit?
5. **(10 points)** Assume that the general purpose register (GPR) of a CPU has 8 registers mainly R1, R2..R8 and three pipeline stages. Write the instruction stages for the instruction- ADD R0, R4, R5.
6. **(10 points)** Intel and AMD use IA64 architecture for their 64bit processors which are CISC type. On the other hand, STMElectronics uses ARM architecture for their Cortex M series processor which is RISC type. What are the differences between RISC and CISC?
7. **(10 points)** What are the advantages of Von Neumann Architecture over Harvard Architecture?
8. **(10 points)** In a 32bit processor, both instruction and data consist of 32 bits. Can an instruction in memory (in binary format) look exactly like a data? Explain.
9. **(10 points)** ARM cortex has a barrel shifter connected to Operand 2. Explain how it will be used for the following instruction: SUBEQ a1,v7,r2,LSR #4
10. **(10 points)** Show one or multiple instructions that can perform a multiplication by 50. You can't use any multiplication instruction such as MUL, S/UMULL or MLA, S/UMLAL